

Impact of Increased Policing on Crime

Introduction

56 of 72 major cities across the United States spend more than 20% of their budgets on police operations (Vera n.d). In the wake of the 2020 protests, several cities - including Minneapolis, Austin, and Los Angeles- redirected portions of their police budgets toward social services. The Harvard Law Review makes an argument for abolishing police on the premise that they “suppress marginalized populations” and that “policing does not prevent harm” (Harvard Law Review 2023). In response to these concerns, an abolitionist sentiment has been growing across the United States in favor of reallocating police spending to public programs that would reduce the root causes of crime and potentially mitigate the effects of systemic inequality. Understanding whether policing is effective is essential for determining the best allocation of our resources as taxpayers, and the debate ultimately raises an important question: Does additional police presence actually reduce crime? If we get an answer “yes” - that indicates the amount of government funding on the public - safety cost should be weighed against the social benefits of reinvestment. If we get the answer “no”, it indicates that additional police presence impacts social benefits in negative ways. There are even stronger arguments for redirecting these funds toward upstream interventions. We propose a research design aimed at assessing the impact of police presence on crime, given that police deployment is primarily driven by event security needs rather than local crime conditions.

Literature Review

Before determining which methodology to use, we did a review of different studies and academic literature on how police presence affects crime. The consensus was that the police elasticity on crime was

negative and local, in the range of approximately -0.3 to -0.7 (Kim et al. 2024). These results suggest that, on average, additional police presence does reduce crime, though the magnitude varies across sources. However, estimating this relationship is challenging methodologically due to the endogeneity of police deployment, due to the reciprocal nature of the relationship between police presence and crime: as more crime happens, more police are deployed, and as more police are deployed, less crime happens.

One approach to estimating this effect is using a regression discontinuity design. A source from a group of researchers at UPenn used geographic RDD, where they expanded the border of UPenn police coverage and estimated the change in crime around the new border, finding that the crime was significantly reduced in that vicinity (Macdonald et al. 2012). Another approach is instrumental variables, such as variation in officer deployment driven by “outside calls” not related to crime, to estimate the causal effect of police removal from specific areas.

A third group makes use of difference-in-differences using panel data across cities, or other natural experiments. The National Bureau of Economic Research and Journal of Criminal Justice have each published papers using dynamic DiD to identify this effect, while the American Economic Review estimates a DiD effect using the event of the London Terror Attack as an exogenous shock. More broadly, these sources validate that dynamic difference-in-differences is commonly used to identify how police presence affects crime rates.

In addition to the information above, we learn some important things about spillovers and heterogeneous effects from the literature. Firstly, the NBER informs us that, if there were a spillover from police displacement, it would bias the estimates towards zero. This tells us that if police officers leave other areas to treat crime, the reduced police presence there and their effect on crime would, in any case, not cause us to overestimate the effect in question. Furthermore, the Journal of Criminal Justice informs us that police presence affects different types of crimes heterogeneously, and the strongest effects were towards violent crimes. This suggests that the choice of crime type would affect our estimate.

Ideal Experiment

In an ideal controlled and randomized trial, we would randomly assign varying levels of police presence across different neighborhoods in Boston on a random day. Neighborhoods would be assigned “high,” “low,” and “standard” policing days, then the violent crime outcome could be measured across the different groups over time. Due to the random assignment of the treatment, differences in violent crime could be attributed to increased police presence.

Our real-world setting introduces a couple of complications, but the design of our study explains them. The biggest challenge is that police presence is endogenous – police officers are directed towards higher crime areas. Our DiD design gets rid of that by using the Boston Marathon as a source of variation in policing, which breaks the reciprocal relationship between crime and police presence. A second concern would be the omitted variable bias from differences in the neighborhoods on and off the marathon route. We address this through our parallel trends assumption, which uses Legacy System data to prove that the neighborhoods off and on the route are similar before the marathon in violent crime. Finally, spillovers are a third complication we had to work around. Marathon policing is typically state and federal troopers, and some additional event security. Meaning, officers from surrounding neighborhoods are confined to their area. So while we aren’t able to entirely randomize our experiment as we would want, we are able to work around these sources of variability.

Threats to Identification

In reality, there are three characteristics of the identification threats that might affect us to complete the real research. Endogeneity of deployment is the first threat in this research. In most large U.S. cities, police deployment is directly driven by the crime data. CompStat or analogous data-driven management systems reallocate patrol resources weekly based on crime hotspots (Weisburd, D 2005). It indicates that there is a strong

direct relationship between police rate and the crime rate, where police are deployed where crime is high, which will result in reducing crime there. In the article “Using Electoral Cycles in Police Hiring to Estimate the Effect of Police on Crime”, it was found that “18 cities found either no relationship or a positive correlation. Examining a panel of 59 larger U.S cities over the period 1970-1992, it is first demonstrated that, as in previous studies, positive cross-city correlations between police and crime emerge” (Levitt, Steven D. 1997). These examples show the extent that endogeneity has been well documented.

The second threat is that we have significantly non-crime characteristic differences in the treated group and the control group, which are all strong predictors or indicators of crime. In terms of the relationship between poor rate and the violent rate, there is a relevant literature that has already confirmed that it illustrates a strong correlation between poverty concentration and violent crime. Robert J. Sampson mentioned his point of view in the paper of Urban Black violence: the effect of male joblessness..., there are few factors all will affect the crime rate independently, those factors including the male population ratio in age with the range from 15 to 25 years old; homeownership rates; mixed-use built environments. (Sampson, Robert J. 1987). The risk of not solving the threat is that any simple easy comparison between the treated group and the control groups will confound the “Marathon Policing effect” with the “Composition effect”.

Spillovers are the last threats for this research. SUTVA requires that the treated group applied to the treated units does not affect the outcomes of the control units. There are three mechanisms that might violate this assumption. The first mechanism is the crime displacement, if increased policing in the treated area drives offenders to commit crimes in neighboring control areas, crime rates in the control areas will rise, which will lead to overestimated the treatment effect. Second, the Police force displacement, in contrast, might also be overestimated if we have more police force from the other districts to support the marathon, so the local police force will decrease. Last but not least, we might underestimate the treatment effect, which means that the result

might shift the estimate toward 0. Because the deterrent effects of policing may spill over into neighboring areas, causing crime to decrease in the control zones as well.

Methodology

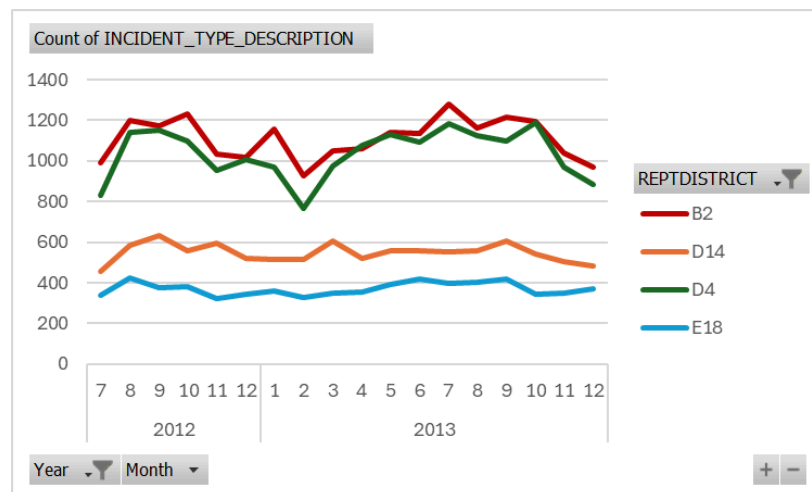
To examine this causal relationship, it is necessary to identify a scenario in which policing is not determined by crime. That is, the analysis requires finding a scenario where we can be certain there is no reciprocal effect to bias our estimation. We use the Boston Marathon as this setting because it is locally relevant, and there is standardized policing across the path, independent of underlying neighborhood crime rates, which controls for the confounding reciprocal effect.

Additionally, we chose to look at violent crime specifically because it is unaffected by reporting differences, enforcement priorities, and police discretion, and because they have a larger social significance (Kelling, George 1999). Therefore, a focus on violent crime will have less measurement error and be less biased. We also wanted our estimation to be unbiased by institutional and event-based differences that occurred around the time of the Boston Marathon bombing and the COVID-19 quarantines, which motivated identifying a time period of 2014-2019, between the two events. Ultimately, our research question was: What is the effect of the sustained post-2013 increase in police presence along the Boston Marathon on violent crime rates in the route-adjacent Boston police district from 2014 to 2019?

To answer that question, we now explicitly identify a hypothesis and a quasi-experimental design to use. The alternative hypothesis is that the increase in police presence from 2014 to 2019 in the Boston Marathon path decreased violent crime, and the null hypothesis is therefore that the increase in police presence during that period at the Boston Marathon did not affect violent crime rates. Our treated group is the Boston neighborhoods on the marathon path, and the control group is the Boston neighborhoods off of the marathon path. The treatment is the increased police presence along the path of the Boston Marathon.

The design we decided to be the best fit was Difference-in-Differences. With the exception of Dynamic DiD, which will be explained in more detail later, the other econometric methods are not sufficient to show causality. A basic OLS is biased because police presence is endogenous, because areas with more crime will have more police officers to begin with. For the IV regression, we were unable to find an instrument that satisfies the exclusion restriction. Instruments like route distance and street closures due to the marathon affect crowd density and crime opportunities directly, meaning they affect both police presence and the outcome variable at the same time. The regression discontinuity design doesn't work because we need a clear cutoff where the treatment changes (e.g., distance to the route) or police presence (varies by different towns on the route, having bigger or smaller crowds, proximity to the finish line).

For our Difference-in-Differences estimate to be causally accepted, two assumptions must hold: the Parallel Trends Assumption and the Stable Unit Treatment Value Assumption (SUTVA). The parallel trends assumption requires that prior to the treatment period, the treated and controlled neighborhoods would have followed the same patterns of violent crime rates over time. Using the Legacy System data from 2012-2013, we can see that the monthly trends for the treated districts are pretty parallel to the control districts pre-2014.



Model 1: Trends in Violent Crime in the Month Prior to Treatment in Back Bay & Brighton versus Roxbury & Hyde Park, 2012-2013

The treated districts we are looking at are D4 and D14, which are Back Bay and Brighton, respectively, and the control districts are B2 and E18, which are Roxbury and Hyde Park, off the marathon route. This allows us to assume that the post-treatment differences in violent crime on and off the marathon route can be attributed to the police presence due to the marathon, rather than pre-existing differences in the neighborhoods, satisfying the parallel trends assumption.

The SUTVA, additionally, requires both heterogeneity in treatment and that the treatment trajectory of one unit doesn't impact the outcomes of another. Marathon policing is standardized year over year, especially after the 2013 bombing and before COVID-19, which makes the treatment consistent over our time period, 2014-2019. The second part of this condition is ensuring there are no spillovers. To mitigate spillovers of crime or police presence to the directly adjacent areas, we selected control districts that are not immediately adjacent but are still in the surrounding area and share pre-treatment characteristics with the treatment groups. Our study treats "treatment" as an institutional regime in place after 2013, rather than a deployment on one day. This regime remained stable throughout the entire study period from 2014 to 2019, satisfying the requirement that treatment intensity remain constant.

Additionally, as stated in the NBER article, spillovers would bias our estimates towards zero, which still results in a significant association, allowing us to still either reject or fail to reject our null hypothesis, if the association is either positive or negative. We were able to mitigate the spillover effect because police in non-marathon towns do not leave their jurisdiction to go on the route if crimes are reported. Policing on the marathon route is handled by federal and state officers, with additional event security to manage the crime on the route so local officers aren't needed. Vice versa because these event staff and federal officers are here specifically for the marathon, they wouldn't be leaving their post for any crime being committed in a local town.

But to reduce even the possibility of spillovers affecting our results, a future direction would be to exclude neighborhoods directly adjacent to the marathon path towns, using a certain distance threshold to create

a buffer zone. If our estimates are similar after removing potentially contaminated neighborhoods, we can be more confident that spillovers are not changing our results.

While we chose a standard DiD, if the parallel trends assumption was not satisfied and the treated and control groups differ in violent crime rates before the marathon, we could use a dynamic DiD model. Our model would look something like:

$$Y_{it} = \alpha + \sum_{k \neq 1} \beta_k * (treated * 1[t = k]) + \gamma_i + \delta_t + \xi_{it}$$

where k indexes years relevant to the treatment period, γ_i are neighborhood fixed effects, δ_t are the year fixed effects, and β_k coefficients determine the treatment effect dynamically over time. Regardless of the dynamic vs standard model, spillovers would bias our estimate towards 0 and would not eliminate the bias. So, since we are certain the parallel trends assumption holds, we can use the standard DiD model, which looks like this:

$$Y_{it} = \alpha + \beta * (treated_i * post_t) + \gamma_i + \delta_t + \xi_{it}$$

where $treated_i$ indicates whether the neighborhood i lies on the marathon path, $post_t$ indicates the post-treatment period, γ_i are neighborhood fixed effects, δ_t are the year fixed effects, and β captures the average effect of police presence due to the marathon on violent crime rates. Regarding the treatment, We identify the treatment period as the sustained, year-over-year increase in police presence in communities along the Boston marathon route following the 2013 Boston Marathon bombings. We define the pre-treatment period as years from 2012 to 2013, Post treatment period is years from 2014 to 2019.

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